

L1b: Time Value of Money, Abstract Assets, and Net Present Value

CHEME 5660 Cornell University

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Objectives for Today

Today we compare **cash flows at different dates**, derive accumulation and discount factors, and use them to value an **abstract asset**.

Today's concepts

- **The time value of money**: cash flows at different dates cannot be added until they are expressed at the same valuation date.
- **An abstract asset**: a sequence of dated signed cash flows. The same representation covers a bond, a stock, a factory, or a car.
- **Net present value**: discount each signed cash flow to the valuation date, then add. Its sign is a statement about the rate you declared, not a guarantee of profit.

Companion notebook: [L1b lecture notebook](#)

Lectures and Examples

LECTURE NOTEBOOK

CHEME-5660-L1b-Lecture-TimeValueMoney-Fall-2026.ipynb

Derive accumulation and discount factors, represent an asset as dated signed cash flows, and compute NPV at a declared valuation rate.

WORKED EXAMPLE

CHEME-5660-L1b-NetPresentValue-Fall-2026-WorkedExample.ipynb

Represent ten years of Tesla Model S ownership as dated cash flows, discount them to time 0, and compute their NPV.

Link: <https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git>

Time Value of Money

A dollar today and a dollar at a future date are **not directly comparable**. Before we compare or add them, we must express both at the same valuation date using a stated conversion rule.

Let P be an amount held at time 0, and let y be a quoted **nominal annual yield** used for valuation. With one compounding interval per year, the amount after one year is given by:

$$F_1 = \underbrace{(1 + y)}_{\mathcal{D}_{1,0}(y;1)} P.$$

For $y > 0$ the factor $\mathcal{D}_{1,0}(y; 1) > 1$ moves time-0 dollars forward and is the **forward accumulation factor**; its inverse $\mathcal{D}_{1,0}^{-1}(y; 1) < 1$ moves future dollars back and is the **discount factor**.

Careful: a later dollar is not *automatically* worth less. That depends on the rate y you declare.

Discrete Compounding

Let $j = 0, 1, 2, \dots$ index the **compounding periods**, and let $i_\ell > -1$ be the **effective rate** over the interval $\ell \rightarrow \ell + 1$. An amount P held at time 0 becomes:

$$F_j = \underbrace{\left[\prod_{\ell=0}^{j-1} (1 + i_\ell) \right]}_{\mathcal{D}_{j,0}(\mathbf{i})} P.$$

For nominal annual yield y compounded n times per year, $i_\ell = y/n$ and the product becomes:

$$\underbrace{\mathcal{D}_{j,0}(y; n) = \left(1 + \frac{y}{n}\right)^j}_{\text{accumulation}}, \quad \underbrace{\mathcal{D}_{j,0}^{-1}(y; n) = \left(1 + \frac{y}{n}\right)^{-j}}_{\text{discount}}.$$

Careful: a flat y is an *assumption*. In L2b, y varies with the date.

Continuous Compounding

Let $g(t)$ be a **continuously compounded growth rate**, carrying units of inverse time, and let T be the horizon. The forward accumulation factor over $0 \rightarrow T$ is given by:

$$\mathcal{D}_{T,0}(g) = \exp \left(\int_0^T g(u) du \right),$$

which for constant g collapses to:

$$\underbrace{\mathcal{D}_{T,0}(g) = e^{gT}}_{\text{accumulation}}, \quad \underbrace{\mathcal{D}_{T,0}^{-1}(g) = e^{-gT}}_{\text{discount}}.$$

A nominal annual yield y compounded n times per year has the continuously compounded equivalent $g_y := n \log(1 + y/n)$, so $(1 + y/n)^{nT} = e^{g_y T}$: these are **two representations of one accumulation rule**, not two different rules.

Log return: the dimensionless product $r = (\Delta t) g$. L2a and L3a build on it.

Who Sets the Valuation Rate?

A valuation rate is **not universal**. It depends on the cash flow, market convention, horizon, and the risks being modeled: no single policy rate prices every asset.

THE FEDERAL RESERVE

- Steers short-term rates through **interest on reserve balances** (IORB), supported by the overnight reverse repo facility; open market operations manage the supply of reserves.
- The **FOMC** sets a target range for the federal funds rate, the overnight rate on unsecured reserve balances traded between eligible institutions.
- Other market rates respond to policy and market conditions, but are **not mechanically pegged** to the federal funds rate.

Link: [FOMC calendar and meeting materials](#)

What Valuation Rate Should I Use?

There is no universal answer, and this lecture does not need one: here the yield y is a **declared model input**. Choosing it well means matching the rate to the cash flow: same currency, same horizon, and an explicit statement of any adjustment made for credit, liquidity, or risk.

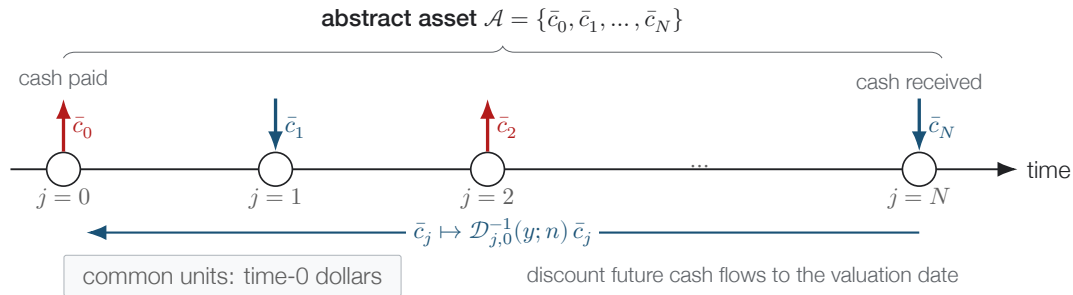
RATE TYPE

A quoted **bond yield** is the rate that matches price to promised cash flows. An **equity required return** estimates future compensation for risk. They are different quantities.

RISK TYPE

Only **systematic risk**, which diversification cannot remove, is expected to earn compensation. More diversifiable risk does not by itself increase expected return. Expected return is not realized return.

An Abstract Asset



Net Cash Flows

An abstract asset is a **sequence of dated cash flows** denominated in a currency. At cash-flow index j , let $\mathbf{c}_j \in \mathbb{R}_{\geq 0}^m$ collect m nonnegative magnitudes and let $\boldsymbol{\nu}_j \in \{-1, +1\}^m$ give their directions, where $+1$ denotes an inflow and -1 an outflow. The **signed net cash flow** at index j is the usual inner product of the magnitude and direction vectors:

$$\bar{c}_j = \langle \mathbf{c}_j, \boldsymbol{\nu}_j \rangle = \sum_{i=1}^m c_{j,i} \nu_{j,i}.$$

Over the dates $j = 0, 1, \dots, N$, the undiscounted total $\sum_{j=0}^N \bar{c}_j$ is **not** a time-0 value: its terms are dollars from different dates.

Next: discount every term to a common date, and the result is net present value.

Net Present Value

Let $\bar{c}_j$ be the **signed net cash flow** on date j , positive for an inflow and negative for an outflow, over the dates $j = 0, 1, \dots, N$. Discount each one to the valuation date with $\mathcal{D}_{j,0}^{-1}(y; n)$, **then add**. The net present value at time 0 is given by:

$$\text{NPV}_0(y) = \sum_{j=0}^N \mathcal{D}_{j,0}^{-1}(y; n) \bar{c}_j = \underbrace{\langle \mathcal{D}_{\star,0}^{-1}(y; n), \bar{\mathbf{c}} \rangle}_{\text{time-0 dollars}}.$$

The $\star$ runs over every cash-flow index, so $\mathcal{D}_{\star,0}^{-1}(y; n)$ and $\bar{\mathbf{c}}$ are ordered $(N+1)$ -vectors and NPV is a single scalar product, which is exactly how we compute it in code.

Worked example: ten years of Model S ownership, $\text{NPV}_0 = -\$121,484$

Summary

This lecture introduced time-value-of-money calculations and the representation of investments as dated cash flows.

Key takeaways

- **Dated dollars must be compared at a common valuation date.** A forward accumulation factor carries value forward and its inverse discounts it back, so cash flows can be added only once they share a date.
- **Rate conventions must be stated.** A nominal yield quoted with a compounding frequency and its continuously compounded equivalent describe one accumulation rule, so the convention travels with the number.
- **NPV values an asset relative to a declared rate.** The sign of the result is a statement about the cash-flow model and the discount rate you declared.

These ideas provide the valuation framework used throughout the course.